AI-POWERED CROP ADVISORY FOR MAHARASHTRA

पीक मित्र

Crop Mitra

Data-driven crop recommendations for the right land, the right season, and better income.

Problem-solving through Data • Scoring • Mobile-first Design

The Problem Farmers Face

Why so many farm decisions go wrong

1

Guessing the wrong crop

Sowing a crop not suited to local soil, rainfall or temperature lowers yield and income every season.

2

No reliable local data

Weather, soil health and market price information is scattered, technical, or hard to access for a village farmer.

3

Market surprises

Producing without knowing the demand or price — good crops, poor or unprofitable selling.

4

Information overload

General advice on the internet is not tuned to the farmer’s tehsil, season or conditions.

Our Solution — Crop Mitra

A mobile-first advisory app built on real district data

📍

Know your location

Pick your district and tehsil — e.g. Latur → Ausa. Everything is tuned to that area.

⚙️

Smart scoring engine

Each crop is scored for soil pH, rainfall, temperature and price fit — out of 100.

🌾

Get ranked advice

See the best crops for your land with suitability %, ranks, and clear Do / Don’t guidance.

🖼️

Learn with pictures

A 55-crop bilingual catalog (Marathi + English) with real images, seasons and cost details.

How It Works

From location to recommendation in 4 steps

1

Select Location

District + Tehsil
(Latur · Ausa)

2

Analyze Data

Soil, rain, temp,
market prices

3

Score Crops

54 crops scored
0–100 per field

4

See Result

Ranked list with
suitability badges

Scores below 60 or rank beyond 20 are marked ‘Not Suitable’ — so farmers never chase the wrong crop.

The Scoring Engine

How a 0–100 suitability score is computed

SOIL pH & TYPE

Healthy soil for the crop = higher base score.

RAINFALL (mm)

Compared with each crop’s ideal rain range.

TEMPERATURE (°C)

Seasonal warmth matched to crop comfort zone.

MARKET PRICE

Profitability signal for the final score.

REGION FIT

Crop history & district-level suitability data.

0–100 SCORE

Weighted average → clean %, easy to understand.

Decision rule: Suitable = Score ≥ 60 AND Rank ≤ 20

If either condition fails, the crop is flagged ‘Not Suitable’ for that tehsil.

Data & Technology Stack

Real data + lightweight modern tools

362	54	52,608	19,548
Regions (districts + tehsils)	Crops (rated & catalogued)	Daily weather records	Crop suitability scores computed

Frontend (mobile-first App)

- React single-page app
- Vite dev server & fast builds
- Bilingual UI (Marathi + English)
- Location picker, catalog, price screens

Backend data (scored offline)

- SQLite DB — regions, crops, soil, weather, prices
- Python scripting — data init & scoring pipeline
- Scores pre-computed → instant app, no server load

Key Features

Designed in the farmer's language — clear and simple

🏠 Bilingual Dashboard

Marathi + English UI with a friendly home screen and easy location change.

📊 Suitability % & Ranks

Every crop shows a clear score, position, and Suitable / Not Suitable tag.

🌱 55-Crop Catalog

Real photos, growing months, yield and cost info for informed choice.

💰 Market Prices

Current price awareness helps the farmer decide what to grow and sell.

🗺️ Tehsil-level Precision

Advice changes with your tehsil — no more one-size-fits-all tips.

📱 Phone-friendly

Works beautifully on a small screen — the device every farmer uses.

Real Example — Latur · Ausa

Top 5 recommended crops on the Ausa advisory screen



Example checks:

Mango (88%) → rank 14, Suitable. Rice (82%) → rank 29, beyond rank 20, flagged Not Suitable for Ausa.

Same crop, different tehsil → different answer. Advice is truly location-specific.

Impact

What Crop Mitra helps achieve

Better crop choices

Matches crop to local soil, rain and temperature — fewer crop failures.

Higher income

Growing what the land and market favour improves yield and returns.

Informed decisions

Clear scores and ranks replace guesswork with confidence.

Sustainable farming

Right crop on right land keeps soil healthy for the long run.

Future Scope

Where Crop Mitra can grow

✓ Live weather & rainfall forecasts wired into scores

✓ Real-time market price APIs for up-to-the-day advice

✓ More crops + regional languages (Hindi, Kannada)

✓ Voice input for elder farmers (Marathi speech)

✓ Crop switches & rotation suggestions for soil health

✓ Offline mode + farmer support helpline in the app

धन्यवाद — Thank You

पीक मित्र — SMARTER FARMING, BETTER LIVELIHOOD

Questions & Feedback Welcome